Celebal Technologies Internship – Week 7 Assignment

Delta Lake MERGE Implementation

Name: Abhishek Raj

Internship: Celebal Technologies Data Engineering Internship

Week: 7

Objective

The objective of this assignment is to perform incremental data processing using **Delta Lake**. The assignment demonstrates how to load data into Delta tables, clean datasets by handling missing values and duplicate records, simulate incremental data, perform MERGE (UPSERT) operations, validate the final dataset, and understand how Delta Lake simplifies modern data engineering workflows.

Dataset Description

Two customer datasets were used in this assignment.

customer_master.csv

The master dataset contains customer information including:

- Customer ID
- First Name
- Last Name
- Email
- Phone Number
- City
- State
- Country
- Signup Date
- Membership

- Status

This dataset intentionally contains duplicate records and missing values to demonstrate data cleaning.

customer_incremental.csv

The incremental dataset contains:

- Existing customers with updated information
- New customer records

This dataset is merged into the master Delta table using Delta Lake MERGE.

Software Requirements

- Python
- Apache Spark (PySpark)
- Delta Lake
- Google Colab (Execution Environment)
- Pandas

Q1. Load Dataset into a Delta Table

Explanation

The master customer dataset was loaded into a Spark DataFrame with automatic schema inference enabled. The DataFrame was then written as a Delta table using Delta Lake storage format.

Screenshot

Q1. Load Dataset into a Delta Table

```python
from pyspark.sql.types import *

master_df = spark.read \
    .option("header", True) \
    .option("inferSchema", True) \
    .csv("customer_master.csv")

master_df.show(5)
```

```
+-----------+----------+---------+--------------------+--------------------+-------+-----------+-------+-----------+----------+--------+
|customer_id|first_name|last_name|               email|               phone|   city|      state|country|signup_date|membership|  status|
+-----------+----------+---------+--------------------+--------------------+-------+-----------+-------+-----------+----------+--------+
|  CUST00001|     Diana|     Bass|paulerica@example...|          9935730990| Mumbai|      Delhi|  India| 2023-10-17|  Platinum|Inactive|
|  CUST00002|  Patricia|     Pena|samanthahoward@ex...|          9489223748|   Pune|      Bihar|  India| 2022-09-08|  Platinum|  Active|
|  CUST00003|    Dalton|    Smith|carolyn94@example...| (913)361-4471x19728| Jaipur|Maharashtra|  India| 2023-11-24|  Platinum|Inactive|
|  CUST00004|      Rose|     Hall|justinadkins@exam...|   (329)594-8104x439|  Delhi|Maharashtra|  India| 2022-02-02|    Silver|  Active|
|  CUST00005|   Phyllis|     Cook|antonio04@example...|001-988-350-0848x...|Kolkata|      Delhi|  India| 2023-06-02|  Platinum|  Active|
+-----------+----------+---------+--------------------+--------------------+-------+-----------+-------+-----------+----------+--------+
only showing top 5 rows
```

```python
master_df.printSchema()
```

```
root
 |-- customer_id: string (nullable = true)
 |-- first_name: string (nullable = true)
 |-- last_name: string (nullable = true)
 |-- email: string (nullable = true)
 |-- phone: string (nullable = true)
 |-- city: string (nullable = true)
 |-- state: string (nullable = true)
 |-- country: string (nullable = true)
 |-- signup_date: date (nullable = true)
 |-- membership: string (nullable = true)
 |-- status: string (nullable = true)
```

```python
delta_master = spark.read \
    .format("delta") \
    .load("/content/customer_master_delta")

delta_master.show(5)
```

```
+-----------+----------+---------+--------------------+--------------------+-------+-----------+-------+-----------+----------+--------+
|customer_id|first_name|last_name|               email|               phone|   city|      state|country|signup_date|membership|  status|
+-----------+----------+---------+--------------------+--------------------+-------+-----------+-------+-----------+----------+--------+
|  CUST00001|     Diana|     Bass|paulerica@example...|          9935730990| Mumbai|      Delhi|  India| 2023-10-17|  Platinum|Inactive|
|  CUST00002|  Patricia|     Pena|samanthahoward@ex...|          9489223748|   Pune|      Bihar|  India| 2022-09-08|  Platinum|  Active|
|  CUST00003|    Dalton|    Smith|carolyn94@example...| (913)361-4471x19728| Jaipur|Maharashtra|  India| 2023-11-24|  Platinum|Inactive|
|  CUST00004|      Rose|     Hall|justinadkins@exam...|   (329)594-8104x439|  Delhi|Maharashtra|  India| 2022-02-02|    Silver|  Active|
|  CUST00005|   Phyllis|     Cook|antonio04@example...|001-988-350-0848x...|Kolkata|      Delhi|  India| 2023-06-02|  Platinum|  Active|
+-----------+----------+---------+--------------------+--------------------+-------+-----------+-------+-----------+----------+--------+
only showing top 5 rows
```

Q2. Handle Missing Values and Remove Duplicate Records

Explanation

Missing values were identified using Spark SQL functions. Records containing null values were removed using dropna(), and duplicate records were removed using dropDuplicates(). The cleaned dataset was then written back to the Delta table.

Screenshot

Q2. Handle Missing Values and Remove Duplicate Records

```
from pyspark.sql.functions import col, count, when

master_df.select([
    count(when(col(c).isNull(), c)).alias(c)
    for c in master_df.columns
]).show()
```

```
+-----------+----------+---------+-----+-----+----+-----+-------+-----------+----------+------+
|customer_id|first_name|last_name|email|phone|city|state|country|signup_date|membership|status|
+-----------+----------+---------+-----+-----+----+-----+-------+-----------+----------+------+
|          0|         0|        0|    0|   80|   0|    0|      0|          0|        50|     0|
+-----------+----------+---------+-----+-----+----+-----+-------+-----------+----------+------+
```

```
clean_df = master_df.dropna()
```

```
clean_df = clean_df.dropDuplicates()
```

```
print("Original Rows:", master_df.count())
print("Clean Rows:", clean_df.count())
```

```
Original Rows: 5030
Clean Rows: 4900
```

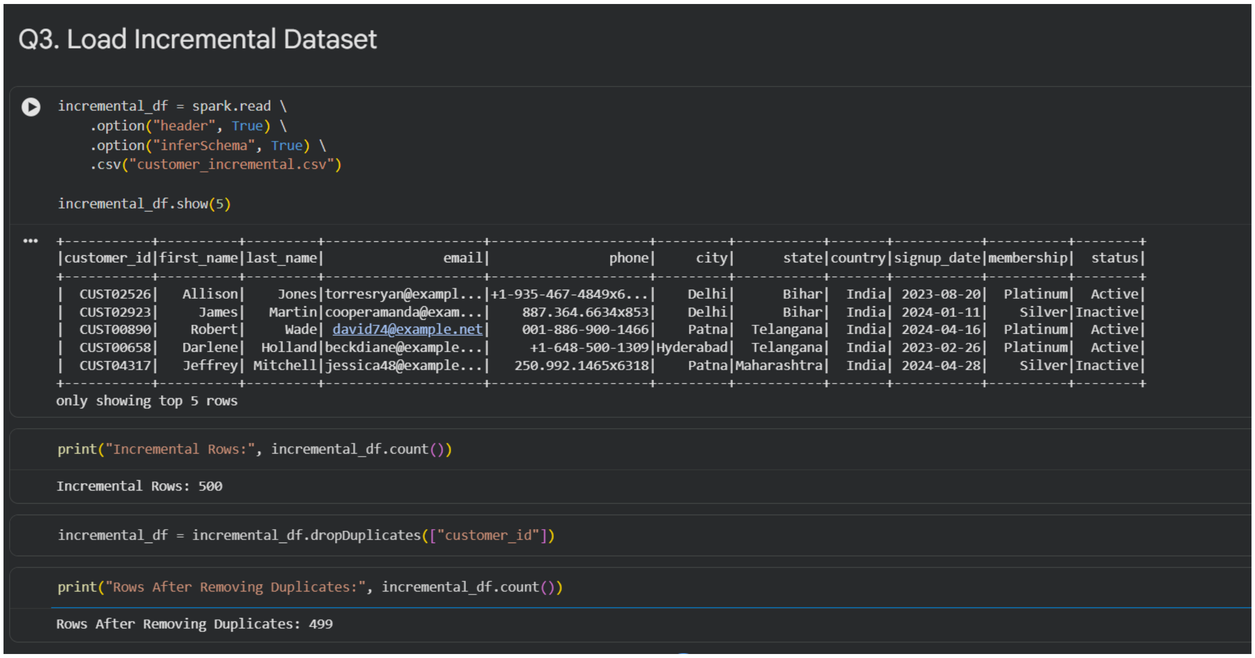
```
clean_df.write \
    .format("delta") \
    .mode("overwrite") \
    .save("/content/customer_master_delta")
```

Q3. Load Incremental Dataset

Explanation

The incremental dataset containing updated and new customer records was loaded into Spark. Duplicate customer IDs were removed to ensure that each customer record participated only once in the MERGE operation.

Screenshot


Q3. Load Incremental Dataset

```
incremental_df = spark.read \
    .option("header", True) \
    .option("inferSchema", True) \
    .csv("customer_incremental.csv")

incremental_df.show(5)
```

```
+-----------+----------+---------+--------------------+--------------------+---------+-----------+-------+-----------+----------+--------+
|customer_id|first_name|last_name|               email|               phone|     city|      state|country|signup_date|membership|  status|
+-----------+----------+---------+--------------------+--------------------+---------+-----------+-------+-----------+----------+--------+
|  CUST02526|   Allison|    Jones|torresryan@exampl...|+1-935-467-4849x6...|    Delhi|      Bihar|  India| 2023-08-20|  Platinum|  Active|
|  CUST02923|     James|   Martin|cooperamanda@exam...|   887.364.6634x853|    Delhi|      Bihar|  India| 2024-01-11|    Silver|Inactive|
|  CUST00890|    Robert|     Wade| david74@example.net|   001-886-900-1466|    Patna|  Telangana|  India| 2024-04-16|  Platinum|  Active|
|  CUST00658|   Darlene|  Holland|beckdiane@example...|     +1-648-500-1309|Hyderabad|  Telangana|  India| 2023-02-26|  Platinum|  Active|
|  CUST04317|   Jeffrey| Mitchell|jessica48@example...|  250.992.1465x6318|    Patna|Maharashtra|  India| 2024-04-28|    Silver|Inactive|
+-----------+----------+---------+--------------------+--------------------+---------+-----------+-------+-----------+----------+--------+
only showing top 5 rows
```

```
print("Incremental Rows:", incremental_df.count())
```

```
Incremental Rows: 500
```

```
incremental_df = incremental_df.dropDuplicates(["customer_id"])
```

```
print("Rows After Removing Duplicates:", incremental_df.count())
```

```
Rows After Removing Duplicates: 499
```

Q4. Apply MERGE Operation

Explanation

The Delta Lake MERGE (UPSERT) operation synchronized the incremental dataset with the master Delta table by updating existing customer records and inserting new records that did not already exist. The MERGE operation performed:

- Updating existing customer records.
- Inserting new customer records.
- Maintaining a single consistent customer record for every customer ID.

Screenshot

Q4. Apply MERGE Operation (Update Existing and Insert New Records)

```python
from delta.tables import DeltaTable

delta_table = DeltaTable.forPath(
    spark,
    "/content/customer_master_delta"
)
```

```python
(
    delta_table.alias("target")
    .merge(
        incremental_df.alias("source"),
        "target.customer_id = source.customer_id"
    )
    .whenMatchedUpdate(
        set={
            "first_name": "source.first_name",
            "last_name": "source.last_name",
            "email": "source.email",
            "phone": "source.phone",
            "city": "source.city",
            "state": "source.state",
            "country": "source.country",
            "signup_date": "source.signup_date",
            "membership": "source.membership",
            "status": "source.status"
        }
    )
    .whenNotMatchedInsert(
        values={
            "customer_id": "source.customer_id",
            "first_name": "source.first_name",
            "last_name": "source.last_name",
            "email": "source.email",
            "phone": "source.phone",
            "city": "source.city",
            "state": "source.state",
            "country": "source.country",
            "signup_date": "source.signup_date",
            "membership": "source.membership",
            "status": "source.status"
        }
    )
    .execute()
)
```

```
DataFrame[num_affected_rows: bigint, num_updated_rows: bigint, num_deleted_rows: bigint, num_inserted_rows: bigint]
```

Q5. Validate Results

Explanation

The final Delta table was validated by checking:

- Final row count
- Duplicate customer IDs
- Dataset summary

Validation confirmed that the MERGE operation completed successfully.

Output Summary

Dataset	Rows
Original Dataset	5030
Clean Dataset	4900
Incremental Dataset	499
Final Dataset	5119
Duplicate Customer IDs	0

The validation confirms that the MERGE operation successfully updated the Delta table while preserving data integrity.

Screenshot

Q5. Validate Results

```
final_df = spark.read \
    .format("delta") \
    .load("/content/customer_master_delta")
```

```
print("Final Row Count:", final_df.count())
```

```
Final Row Count: 5119
```

```
from pyspark.sql.functions import col

duplicate_count = (
    final_df.groupBy("customer_id")
    .count()
    .filter(col("count") > 1)
    .count()
)

print("Duplicate Customer IDs:", duplicate_count)
```

```
Duplicate Customer IDs: 0
```

Q6. Display Final Dataset and Summary

```python
final_df.show(20, truncate=False)
```

customer_id	first_name	last_name	email	phone	city	state	country	signup_date	membership	status
CUST00001	Diana	Bass	paulerica@example.org	9935730990	Mumbai	Delhi	India	2023-10-17	Platinum	Inactive
CUST00002	Patricia	Pena	samanthahoward@example.net	9489223748	Pune	Bihar	India	2022-09-08	Platinum	Active
CUST00003	Dalton	Smith	carolyn94@example.com	(913)361-4471x19728	Jaipur	Maharashtra	India	2023-11-24	Platinum	Inactive
CUST00004	Rose	Hall	justinadkins@example.com	(329)594-8104x439	Delhi	Maharashtra	India	2022-02-02	Silver	Active
CUST00005	Phyllis	Cook	antonio04@example.net	001-988-350-0848x43155	Kolkata	Delhi	India	2023-06-02	Platinum	Active
CUST00006	Wesley	Martin	emoore@example.com	859-552-2841x0783	Jaipur	Uttar Pradesh	India	2024-01-04	Silver	Inactive
CUST00007	Daniel	Campbell	tterrell@example.net	(579)822-1239x4281	Hyderabad	Delhi	India	2023-08-27	Silver	Inactive
CUST00008	Brandy	Brewer	vanderson@example.net	+1-343-662-7124x976	Hyderabad	Bihar	India	2022-12-15	Silver	Inactive
CUST00009	Edward	Love	bishopeugene@example.org	737-303-1192x667	Delhi	Delhi	India	2022-04-15	Silver	Active
CUST00010	Hunter	Johnson	andrewyoung@example.com	836.828.0929x38037	Chennai	Telangana	India	2024-05-17	Silver	Inactive
CUST00011	William	Johnson	john80@example.com	+1-664-309-1475x815	Mumbai	Uttar Pradesh	India	2023-07-04	Silver	Inactive
CUST00012	Sheila	Martin	melissa18@example.net	517.821.0574	Kolkata	Karnataka	India	2024-04-29	Platinum	Active
CUST00013	Rebecca	Hill	wrightsarah@example.net	5273704592	Mumbai	Delhi	India	2023-12-23	Platinum	Active
CUST00014	Richard	Davis	christopherboyd@example.net	824.298.4604x50411	Hyderabad	Maharashtra	India	2024-03-02	Silver	Active
CUST00015	Jennifer	Jimenez	susan80@example.net	998-728-6546	Hyderabad	Rajasthan	India	2023-01-25	Platinum	Inactive
CUST00016	Jonathan	Harper	joellewis@example.org	459.739.7629x137	Chennai	Karnataka	India	2022-06-16	Silver	Inactive
CUST00017	Sonia	Young	joe68@example.com	218.287.8749	Mumbai	Bihar	India	2023-12-20	Platinum	Active
CUST00018	Kathleen	Mitchell	kennethclark@example.org	(525)298-4346	Lucknow	Uttar Pradesh	India	2022-06-17	Gold	Active
CUST00019	April	Frederick	julianadams@example.com	(939)663-7638x01698	Chennai	Delhi	India	2023-12-03	Silver	Active
CUST00020	Eileen	Cunningham	malonelinda@example.net	784-963-1883x364	Chennai	Uttar Pradesh	India	2024-04-04	Gold	Active

only showing top 20 rows

```python
final_df.printSchema()
```

```
root
 |-- customer_id: string (nullable = true)
 |-- first_name: string (nullable = true)
 |-- last_name: string (nullable = true)
 |-- email: string (nullable = true)
 |-- phone: string (nullable = true)
 |-- city: string (nullable = true)
 |-- state: string (nullable = true)
 |-- country: string (nullable = true)
 |-- signup_date: date (nullable = true)
 |-- membership: string (nullable = true)
 |-- status: string (nullable = true)
```

```python
print("Original Dataset :", master_df.count())
print("Clean Dataset :", clean_df.count())
print("Incremental Dataset :", incremental_df.count())
print("Final Dataset :", final_df.count())
```

```
Original Dataset : 5030
Clean Dataset : 4900
Incremental Dataset : 499
Final Dataset : 5119
```

Conclusion

This assignment successfully demonstrated incremental data processing using Delta Lake. The master customer dataset was cleaned by removing missing values and duplicate records before being stored as a Delta table. An incremental dataset containing updated and new customer records was then merged into the Delta table using the MERGE operation. Finally, the merged dataset was validated to ensure correct row counts and the absence of duplicate customer IDs. This assignment provided practical experience with Delta Lake's ACID transactions, efficient UPSERT operations, and incremental data processing techniques commonly used in modern data engineering pipelines.

References

- Apache Spark Documentation
- Delta Lake Documentation
- PySpark API Documentation
- Celebal Technologies Data Engineering Internship Material